



MANTLE SENTINEL

Behavioral Anomaly Detection for Smart Contracts

Hackathon Track: AI Alpha & Data

The Problem

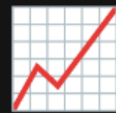
\$3.4B stolen in crypto hacks in 2025

- ▶   Signature-based tools miss novel attacks
- ▶   LLM monitors are slow and non-deterministic
- ▶  New protocols have **zero historical data** to train on

Our Approach

Every contract has a behavioral fingerprint.

Sentinel detects the moment it changes.



Normal behavioral pattern



Anomalous drift detected

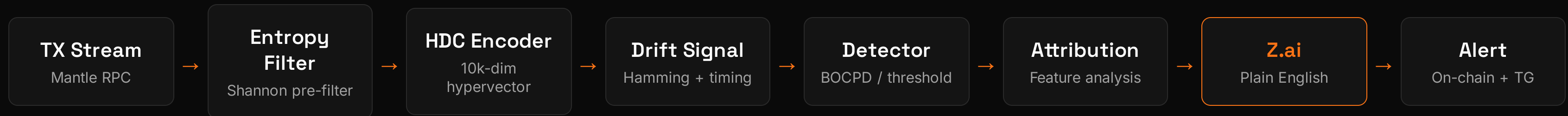


10,000 dimensions

0 training

0 GPU

Pipeline



⚡ LLM is NEVER in the detection loop — detection is deterministic and reproducible, byte for byte.

Z.ai Integration

STRUCTURED FINDING (input)

feature: gas_p99

drift: 0.847

baseline: 142,000

current: 1,203,000

window: 45-52

↓ Z.ai GLM ↓

PLAIN ENGLISH (output)

"Gas consumption spiked 8.5× above baseline in windows 45–52, consistent with a complex multi-call pattern not seen in the warmup period."

- ▶ Strict template: may only restate Tier-5 findings
- ▶ No hallucinated severity
- ▶ No invented causes
- ▶ Without API key → deterministic fallback (CI-safe)

Live Scan Results

CONTRACT	HEALTH	STATUS
USDC.e	83 /100	Healthy ✓
Lendle	81 /100	Healthy ✓
WMNT	74 /100	Moderate ⚠
USDT	70 /100	Moderate ⚠
mETH	68 /100	Moderate ⚠

129 tests

0 false positives

4.3× separation ratio

On-Chain Alert Anchoring

SentinelAlertRegistry

0x0899E1507CFfeF8620455721F5bd528Bb072187

Mantle Mainnet · Chain ID 5000



Immutable



Verifiable



Auditable on Mantlescan

Developer Experience

```
$ python -m sentinel scan 0x09bc4e... --min-health 60
```

Health Score: 83 / 100 

Drift (median): 0.221

Alerts: 4

One command. Full behavioral profile.

GitHub Actions CI gate

Pre-commit hooks

Telegram alerts 

Roadmap

NOW

- ▶ Single-contract monitoring
- ▶ CLI interface
- ▶ Telegram alerts
- ▶ On-chain anchoring

NEXT

- ▶ Multi-contract monitoring
- ▶ Python SDK
- ▶ Confidence scoring
- ▶ Alert tiers

FUTURE

- ▶ EVM multi-chain
- ▶ SaaS dashboard
- ▶ Insurance oracle
- ▶ MCP integration



MANTLE SENTINEL

Detect Before They Hit



mntsentinel.xyz



mntsentinel.xyz/dashboard



github.com/alexbelij/mantle-sentinel



[@MantleSentinelBot](https://t.me/MantleSentinelBot)